

Point-Vortex Collapse Without Rotation: A Cluster Mechanism, a Phase Diagram and a Continuum Limit

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Abstract

In a self-similar collapse of point vortices the configuration turns through the angle P while the square of its size decreases by the factor e . A companion paper proved that $P > \sqrt{3}/2$ for three Euler vortices, that the same constant is the limit for a strong vortex carrying weak tight pairs of opposite sign, and, with computer-assisted proofs, that eleven vortices in the $\alpha = 2$ model and sixty in the surface quasi-geostrophic (SQG) model can collapse without rotating at all. We first show that the constant $\sqrt{3}/2$ belongs to a larger class: a strong Euler vortex carrying any number of weak tight clusters, of any sizes and signs, has $P \geq \sqrt{3}/2 - o(1)$ as the clusters weaken. If P stays bounded, every cluster is close to a translating relative equilibrium whose net circulation is half the sum of the squares of its circulations, measured in units of the strong one; a single isolated weak vortex forces P to grow like the inverse of the weak circulations, with an explicit constant. In particular no such configuration collapses without rotation, whatever the number of vortices. A formal next-order expansion, confirmed by exact collapses computed at fifty digits, shows that triples of signs $(+, +, -)$ beside a positive strong vortex go below $\sqrt{3}/2$, which is why four vortices are the first to do so. We then map, numerically, the least α at which N vortices can collapse without rotation along two families: the least N is 5, 6, 8, 11, 17, 29 and 60 for $\alpha = 6, 4, 3, 2, 1.5, 1.2$ and 1, and the thresholds extrapolate to a positive limit between 0.66 and 0.80, not to the Euler value 0. No self-similar collapse with nonzero total circulation is mirror symmetric; numerically, the two certified collapses without rotation are linearly unstable, with 8 and 57 unstable modes, and perturbed motions leave them without turning. Finally, the Euler minimizers of the companion paper approach a continuum of two point vortices and a positive vortex sheet, whose least winding we compute to forty digits, $P_\infty = 0.47736353369161202484\dots$

Keywords: point vortices; vortex collapse; self-similar motion; relative equilibria; surface quasi-geostrophic equation; vortex sheet.

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1 Introduction

Three point vortices can collapse onto a point in finite time, keeping their shape while they shrink and turn; this was found by Gröbli (see [1, p. 17, Fig. 3] for the history), and the similarity solution for any number of vortices was written by Kimura [7, Sect. 3.1] and, with explicit examples, by Demina and Kudryashov [3]. In such a collapse every vortex runs a logarithmic spiral about the collision point, and the number P , the angle through which the configuration turns while the square of its size decreases by the factor e , measures how tightly the spirals wind. The companion paper

[6] studies the least value of P . For three Euler vortices it proves $P > \sqrt{3}/2$, sharply, and in the α -models, where a vortex induces the velocity $\Gamma r^{-\alpha-1}/(2\pi)$, it proves $P > \sqrt{3+\alpha}/(2+\alpha)$ for three vortices; it shows that the constant $\sqrt{3}/2$ is also the limit for a strong vortex carrying weak, tight pairs of opposite sign [6, Theorem 3]; and with computer-assisted proofs it shows that the bound fails for four or more Euler vortices and that eleven vortices in the $\alpha = 2$ model and sixty in the SQG model ($\alpha = 1$) can collapse without rotating, every vortex running straight into the collision point [6, Theorems 4, 5].

This paper takes up three questions that the companion paper leaves open.

Why is $\sqrt{3}/2$ the limit for so many configurations, and when can it be beaten? In Section 3 we replace the pairs of [6, Theorem 3] by weak tight clusters of any size and signs (Theorem 1). The bound $\sqrt{3}/2 - o(1)$ survives, and the proof shows what a cluster must look like in a collapse with bounded P : a translating relative equilibrium of the cluster's own vortices, carrying the net circulation $(1/(2\Gamma_0)) \sum_k \Gamma_k^2$, whose self-induced translation makes an angle of 60° , at equal speed, with the motion induced by the strong vortex when P is near its infimum. Consequently no configuration in this regime collapses without rotation, for any number of vortices (Corollary 1). A formal next-order expansion, which we confirm numerically but do not prove, explains why the infimum can be undershot as soon as a cluster has three vortices, and hence why four vortices are the first to go below $\sqrt{3}/2$.

Where do collapses without rotation live? In Section 4 we compute, numerically, the least α at which N vortices can collapse without rotation along two families, for N up to 73 and 79. The thresholds decrease with N but extrapolate to a positive limit, between 0.66 and 0.80 depending on the fit; fits that force the Euler value 0 are much worse. This is evidence, along these families only, that Euler vortices do not collapse without rotation.

What do these collapses look like, and are they robust? In Section 5 we prove that no self-similar collapse with nonzero total circulation is mirror symmetric (Proposition 1), describe numerically the nine-parameter family of collapses without rotation of eleven vortices at $\alpha = 2$, and show numerically that both certified collapses without rotation are linearly unstable, with perturbed motions that leave the collapse without having turned. In Section 6 we compute the continuum limit of the Euler minimizers of [6, Sect. 7]: two point vortices and a positive vortex sheet in self-similar collapse, whose least winding agrees with the extrapolation of the finite family. Section 7 lists the programs that check every statement, Section 8 discusses the literature, and Section 9 lists open problems.

Theorem 1, Corollary 1 and Proposition 1 are proved. Everything in Sections 4, 5.2, 5.3 and 6, and the expansion of Section 3.3, is numerical, computed in binary64 or in high-precision arithmetic but not in interval arithmetic, and is labelled as such.

2 Setting

2.1 Self-similar collapse and the winding number

In the α -models point vortices at positions $z_j \in \mathbb{C}$ with real circulations Γ_j move by

$$\frac{dz_j}{dt} = \frac{i}{2\pi} \sum_{k \neq j} \Gamma_k \frac{z_j - z_k}{|z_j - z_k|^{\alpha+2}}, \quad (1)$$

so that a vortex of circulation Γ induces the speed $|\Gamma| r^{-\alpha-1}/(2\pi)$ at distance r ; $\alpha = 0$ is the Euler equation and $\alpha = 1$ is SQG, and we use the conventions of [6, Sect. 2]. For $\alpha = 0$ we write the law as

$$\bar{\dot{z}}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}. \quad (2)$$

Suppose $\sum_j \Gamma_j \neq 0$ and let $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$ be the center of vorticity. If $\dot{z}_j = \kappa(z_j - z_c)$ for all j at one instant, with $\kappa \in \mathbb{C}$, the motion is self-similar, and it is a collapse when $\text{Re } \kappa < 0$ [6, Sect. 2]; the vortices then collide at z_c at a finite time, every vortex moves on a logarithmic spiral about z_c , and the number

$$P = \frac{|\text{Im } \kappa|}{-2 \text{Re } \kappa} \quad (3)$$

is the angle through which the configuration turns while the square of its size decreases by the factor e [6, Eq. (4)]. A collapse is *without rotation* when κ is real, that is, $P = 0$: then every vortex moves on a straight line into the collision point. For the Euler law it is convenient to write

$$\Lambda = 2\pi i \bar{\kappa}, \quad \text{Re } \Lambda = 2\pi \text{Im } \kappa, \quad \text{Im } \Lambda = 2\pi \text{Re } \kappa, \quad P = \frac{|\text{Re } \Lambda|}{-2 \text{Im } \Lambda}, \quad (4)$$

so that a self-similar Euler configuration satisfies $\sum_{k \neq j} \Gamma_k / (z_j - z_k) = \Lambda \overline{(z_j - z_c)}$ for every j , and it collapses when $\text{Im } \Lambda < 0$.

Two classical necessary conditions constrain every self-similar collapse: the angular impulse $\sum_j \Gamma_j |z_j - z_c|^2$ is conserved and proportional to the square of the size, so it vanishes; and the Hamiltonian is conserved while it scales with the size, which for $\alpha = 0$ forces $\sum_{j < k} \Gamma_j \Gamma_k = 0$ and for $\alpha \neq 0$ forces $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^{-\alpha} = 0$ [6, Sect. 2, Lemma 1, Sect. 7], [3]. Once translations, rotations and the scalings of lengths and circulations are divided out, the self-similar collapses of N vortices form, where their equations are independent, a manifold of dimension $N - 1$ [6, Sect. 7], and those without rotation a manifold of dimension $N - 2$ at fixed α , or $N - 1$ when α is free.

2.2 What the companion paper proves

For three Euler vortices every self-similar collapse has $P > \sqrt{3}/2$, and the constant is approached as one circulation tends to zero [6, Theorem 1, Corollary 1]; for three vortices in the α -models, $P > \sqrt{3 + \alpha}/(2 + \alpha)$ for every $\alpha > -2$ [6, Theorem 2], so three vortices never collapse without rotation. For a strong vortex carrying m weak, tight pairs of opposite sign, with pair circulations of order γ , $P \geq \sqrt{3}/2 - o(1)$ as $\gamma \rightarrow 0$, and P is close to $\sqrt{3}/2$ only when every pair is tilted by 60° from the direction away from the strong vortex [6, Theorem 3]; for γ below a threshold that is not explicit the bound is strict, $P \geq \sqrt{3}/2 + (\sqrt{3}/8)c^2\gamma^2$ [6, Proposition 4]. Computer-assisted proofs show that four, five and six Euler vortices collapse with $P < \sqrt{3}/2$, with strict local minima $0.7978967838\dots$, $0.7448144569\dots$ and $0.7136801485\dots$ [6, Theorem 4], and that eleven vortices at $\alpha = 2$ and sixty SQG vortices collapse without rotation, in families of dimension 9 and 58 [6, Theorem 5]. Numerically, a two-arm family of Euler minimizers reaches $P = 0.4793959201\dots$ at $N = 603$ [6, Sect. 7]. No Euler collapse without rotation is known; for $\alpha = 0$ a collapse with κ real is the same as a relative equilibrium of the imaginary circulations $i\Gamma_j$ [6, Sect. 10].

3 A strong vortex with weak tight clusters

In this section the vortices obey the Euler law (2).

3.1 The class and the theorem

Let $n \geq 1$ and $0 < c \leq 1$. The class $\mathcal{K}(n, c)$ consists of the configurations of $1 + n$ point vortices with circulation 1 at the origin and circulations $\Gamma_k = \gamma g_k$ at z_k , $k = 1, \dots, n$, where $\gamma > 0$ and

$c \leq |g_k| \leq 1/c$, the signs of the g_k being arbitrary, together with a partition of $\{1, \dots, n\}$ into clusters C and points $Z_C \in \mathbb{C}$ such that

$$c \leq |Z_C| \leq \frac{1}{c}, \quad |Z_C - Z_D| \geq c \ (C \neq D), \quad |z_k - Z_C| \leq \frac{\gamma}{c} \ (k \in C), \quad |z_k - z_l| \geq c\gamma \ (k \neq l \in C). \quad (5)$$

A cluster may consist of a single vortex. For $k \in C$ we write $\zeta_k = (z_k - Z_C)/\gamma$, so that $|\zeta_k| \leq 1/c$ and $|\zeta_k - \zeta_l| \geq c$, and

$$G_C = \sum_{k \in C} g_k, \quad D_C = \sum_{k \in C} g_k \zeta_k, \quad S_C = \sum_{k \in C} g_k^2, \quad w_k = \sum_{l \in C, l \neq k} \frac{g_l}{\zeta_k - \zeta_l}, \quad q_C = -\frac{2D_C}{S_C Z_C} = p_C + iy_C.$$

Thus γG_C is the net circulation of C and D_C its dipole moment in units of γ^2 , and $w_k/(2\pi i)$ is the conjugate of the velocity that the other vortices of C induce at z_k . We measure the net circulation of a cluster by the dimensionless ratio

$$\hat{\nu}_C = \frac{\sum_{k \in C} \Gamma_k}{\frac{1}{2\Gamma_0} \sum_{k \in C} \Gamma_k^2} = \frac{2G_C}{\gamma S_C}, \quad \Gamma_0 = 1,$$

which does not change when all circulations are multiplied by a positive constant. The normalization is no loss of generality: a translation puts the strong vortex at the origin, a positive factor on the circulations leaves P unchanged, and the map $(\Gamma_j, z_j) \mapsto (-\Gamma_j, \bar{z}_j)$ maps solutions of (2) to solutions, collapses to collapses with the same P [6, Sect. 3.1], and $\mathcal{K}(n, c)$ onto itself.

The class of [6, Theorem 3], a strong vortex with m pairs γa_j at Z_j and $-\gamma b_j$ at W_j with $c \leq a_j, b_j \leq 1/c$, $c \leq |Z_j| \leq 1/c$, $c\gamma|Z_j| \leq |W_j - Z_j| \leq \gamma|Z_j|/c$ and $|Z_j - Z_l| \geq c$, is contained in $\mathcal{K}(2m, c^2)$, with the pairs as clusters and $Z_C = Z_j$; for a pair, $q_C = (W_j - Z_j)/(\gamma a_j Z_j)$ is the quantity u_j of [6].

Theorem 1 (A strong vortex with weak tight clusters). *Let $n \geq 1$ and $0 < c \leq 1$, and consider the configurations of $\mathcal{K}(n, c)$ that collapse self-similarly, $\dot{z}_i = \kappa(z_i - z_c)$ for all $1 + n$ vortices with $\text{Re } \kappa < 0$.*

(a) *For every $\delta > 0$ there is $\gamma_1 > 0$, depending only on n, c and δ , such that every such collapse with $\gamma < \gamma_1$ has $P > \sqrt{3}/2 - \delta$.*

(b) *For every $\delta > 0$ and $\bar{P} > 0$ there is $\gamma_2 > 0$, depending only on n, c, δ and $\bar{P}$, such that every such collapse with $\gamma < \gamma_2$ and $P \leq \bar{P}$ has, for every cluster C : vortices of both signs in C ; $|p_C - 1/2| < \delta$; $|\hat{\nu}_C - 1| < \delta$; $\max_{k, l \in C} |w_k - w_l| < \delta$; $y_C > 0$; and*

$$\left| P - \frac{y_C^2 + 3/4}{2y_C} \right| < \delta, \quad \text{where} \quad \frac{y_C^2 + 3/4}{2y_C} = \frac{\sqrt{3}}{2} + \frac{(y_C - \sqrt{3}/2)^2}{2y_C}.$$

(c) *For every $\delta > 0$ there are $\eta > 0$ and $\gamma_3 > 0$, depending only on n, c and δ , such that every such collapse with $\gamma < \gamma_3$ and $P < \sqrt{3}/2 + \eta$ has $|q_C - e^{i\pi/3}| < \delta$ for every cluster C and $||Z_C|/|Z_D| - 1| < \delta$ for all clusters C and D .*

(d) *If some cluster consists of a single vortex, every such collapse with $\gamma \leq c^3/(20n)$ has $P \geq c^3/(40n\gamma)$.*

For every even $n \geq 2$ the constant $\sqrt{3}/2$ in (a) cannot be increased.

In words: when the clusters are weak, a collapse with bounded winding has every cluster close to a translating relative equilibrium (all the w_k of the cluster nearly equal) whose net circulation, positive, is close to $(1/(2\Gamma_0)) \sum_k \Gamma_k^2$; and P can come close to $\sqrt{3}/2$ only if every q_C is close to $e^{i\pi/3}$ and all clusters are at nearly the same distance from the strong vortex. The self-induced velocity

V_C of a cluster and the velocity V_s induced there by the strong vortex satisfy $V_C/V_s = 1/\sqrt{q_C}$ in the limit (Step 4 below), so $q_C = e^{i\pi/3}$ means that the cluster translates at the speed induced by the strong vortex, at the angle 60° to it. On $p_C = 1/2$, with $t = \arg q_C$, one has $|V_C|/|V_s| = 2 \cos t$ and $P = (2 + \cos 2t)/(2 \sin 2t)$.

Corollary 1. *For every $n \geq 1$ and $0 < c \leq 1$ there is $\gamma_* > 0$ such that no configuration of $\mathcal{K}(n, c)$ with $\gamma < \gamma_*$ collapses self-similarly without rotation. In particular, for no number of vortices is a collapse without rotation possible in the regime of a strong vortex with sufficiently weak tight clusters.*

Proof. Take $\gamma_* = \gamma_1$ of Theorem 1(a) with $\delta = \sqrt{3}/4$; then $P > \sqrt{3}/4 > 0$. $\square$

Remark 1 (What the class contains and excludes). The theorem covers all of $\mathcal{K}(n, c)$: every partition into clusters, and clusters of every shape allowed by (5), singletons included. The proof shows more than the statement. A single vortex forces the stationary branch of Step 4, where $P \rightarrow \infty$, and (d) makes this explicit. A cluster of two or more vortices of one sign cannot occur at all in a collapse once γ is below a constant that depends only on n and c , since in the limit it would need $G_C^2 = S_C$, while $G_C^2 - S_C = 2 \sum_{k < l} g_k g_l > 0$ (Step 4). A cluster that satisfies its own harmonic condition, $\sum_{k < l \in C} g_k g_l = 0$, could collapse by itself; such clusters fall in the stationary branch as well. Two restrictions are part of the hypotheses and cannot be dropped. Nested scales are excluded: a tight pair of size γ^2 inside a cluster of size γ violates $|z_k - z_l| \geq c\gamma$ for every fixed c , and the theorem says nothing about it. And γ_1 is not uniform as $c \rightarrow 0$: the seven-vortex collapse of Demina and Kudryashov [3, Table 1, Fig. 1a], with $P = 12433/(1240\sqrt{155}) = 0.8053 \dots$ [6, Sect. 3.2], lies in $\mathcal{K}(6, c)$ with every weak vortex a singleton and $\gamma = c$, for every $c \leq 1/3$, after its circulations are divided by the central one; so $\gamma_1(6, c, \delta) \leq c$ whenever $\delta < \sqrt{3}/2 - 0.8054$.

Remark 2 (No bound at a fixed circulation). For pairs, $P > \sqrt{3}/2$ at every γ below a threshold [6, Proposition 4]. There is no such statement for clusters: exact collapses of a strong vortex with a single triple of signs $(+, +, -)$ have $P < \sqrt{3}/2$ at $\gamma = 10^{-3}$ and 10^{-4} (Section 3.3 and Table 1; numerical, at fifty digits). So the δ in (a) cannot be removed, and the constant $\sqrt{3}/2$ is an infimum over the class that is undershot at every small γ .

Remark 3 (The net circulation and the classical condition). With one cluster, the necessary condition $\sum_{i < j} \Gamma_i \Gamma_j = 0$ reads $\gamma G_C + \gamma^2(G_C^2 - S_C)/2 = 0$, which gives exactly $\hat{\nu}_C = 1 - \gamma^2 \nu_C^2/S_C$ with $\nu_C = G_C/\gamma$, so $\hat{\nu}_C = 1 - \gamma^2 S_C/4 + O(\gamma^4)$. The exact collapses of Section 3.3 satisfy this to 10^{-36} . With several clusters the classical condition only fixes $\sum_C \nu_C = \sum_C S_C/2 + O(\gamma^2)$; part (b) says that each cluster separately carries the net circulation $\gamma^2 S_C/2$ in the limit.

3.2 Proof of Theorem 1

We follow the proof of [6, Theorem 3]. It suffices to prove three statements about a sequence of collapses in $\mathcal{K}(n, c)$ with $\gamma = \gamma_j \rightarrow 0$, in which we write $P_j, \Lambda_j, q_{C,j}$ and so on: (S1) $\liminf P_j \geq \sqrt{3}/2$; (S2) if $P_j \rightarrow \sqrt{3}/2$, then $q_{C,j} \rightarrow e^{i\pi/3}$ and $|Z_{C,j}|/|Z_{D,j}| \rightarrow 1$ for all clusters C and D ; (S3) if $\sup_j P_j < \infty$, then for all large j every cluster has vortices of both signs, and for every cluster $p_{C,j} \rightarrow 1/2$, $\hat{\nu}_{C,j} \rightarrow 1$, $\max_{k,l \in C} |w_{k,j} - w_{l,j}| \rightarrow 0$, $\liminf_j y_{C,j} > 0$ and $P_j - (y_{C,j}^2 + 3/4)/(2y_{C,j}) \rightarrow 0$. Indeed, if (a) failed for some δ there would be a sequence with $\gamma_j \rightarrow 0$ and $P_j \leq \sqrt{3}/2 - \delta$, contrary to (S1). If (c) failed for some δ , there would be a sequence with $\gamma_j \rightarrow 0$ and $P_j < \sqrt{3}/2 + 1/j$ for which one of the inequalities fails for every j ; after passing to a subsequence the partition into clusters is the same for all j (there are finitely many partitions) and the failing inequality is the same, for the same clusters; by (S1), $P_j \rightarrow \sqrt{3}/2$, contrary to (S2). The same argument with $P_j \leq \bar{P}$ reduces (b) to (S3). Part (d) is proved separately at the end.

Throughout, $\gamma \leq \gamma_0 = c^3/(16n)$, which holds along a sequence with $\gamma_j \rightarrow 0$ after finitely many terms are discarded, and $O(\gamma^k)$ denotes a quantity bounded by γ^k times a constant that depends only on n and c . Each identity used below is a direct computation, verified exactly in `verify_cluster_identities.py`; the remainders are checked in `verify_cluster_remainders.py` (Section 7).

Step 1: a priori bounds. Since $\gamma/c \leq c^2/16$, every z_k with $k \in C$ lies within $c/16$ of Z_C , so $c/2 \leq |z_k| \leq 2/c$, and a vortex of C and one of $D \neq C$ are at distance at least $c - 2\gamma/c \geq c/2$. The total circulation $1 + \gamma \sum_k g_k$ lies in $[1/2, 3/2]$, so $|z_c| \leq 2\gamma \sum_k |g_k| |z_k| \leq 4n\gamma/c^2 \leq c/4$. For $k \in C$ let $E_C(z) = \sum_{l \notin C} \gamma g_l / (z - z_l)$, the field of the weak vortices outside C ; it is holomorphic on the disc $|z - Z_C| \leq c/4$, where every z_l , $l \notin C$, is at distance at least $c/2$, so there $|E_C| \leq 2n\gamma/c^2$, and E'_C and E''_C are also $O(\gamma)$. Moreover $|w_k| \leq n/c^2$. The equation $\dot{z}_k = \kappa(z_k - z_c)$ at a weak vortex $k \in C$ reads, by (2) and (4),

$$\Lambda \overline{(z_k - z_c)} = \frac{1}{z_k} + w_k + E_C(z_k), \quad (6)$$

since the vortices of C other than k contribute $\sum_{l \in C, l \neq k} \gamma g_l / (\gamma(\zeta_k - \zeta_l)) = w_k$. Its right side is at most $2/c + n/c^2 + 2n\gamma/c^2 \leq 4n/c^2$ in absolute value and $|z_k - z_c| \geq c/4$, so $|\Lambda| \leq 16n/c^3$.

Step 2: the weighted sums. Put

$$T_C = \Lambda \overline{(Z_C - z_c)} - \frac{1}{Z_C} - E_C(Z_C).$$

Since $z_k = Z_C + \gamma \zeta_k$, (6) gives

$$w_k - T_C = \gamma \Lambda \overline{\zeta_k} + \left(\frac{1}{Z_C} - \frac{1}{z_k} \right) + (E_C(Z_C) - E_C(z_k)) = O(\gamma), \quad k \in C. \quad (7)$$

The internal field satisfies two classical identities, obtained by symmetrizing the double sums:

$$(I1) \quad \sum_{k \in C} g_k w_k = 0, \quad (I2) \quad \sum_{k \in C} g_k \zeta_k w_k = \frac{1}{2} \sum_{k \neq l} g_k g_l = \frac{G_C^2 - S_C}{2}.$$

Multiply (6) by g_k and sum over $k \in C$. With the exact expansion $\sum_k g_k / z_k = G_C / Z_C - \gamma D_C / Z_C^2 + \gamma^2 \sum_k g_k \zeta_k^2 / (Z_C^2 z_k)$, the Taylor expansion $\sum_k g_k E_C(z_k) = G_C E_C(Z_C) + \gamma D_C E'_C(Z_C) + O(\gamma^3)$ and (I1), this gives

$$G_C T_C = -\gamma \left(\Lambda \overline{D_C} + \frac{D_C}{Z_C^2} \right) + O(\gamma^2). \quad (8)$$

Multiply (6) by $g_k \zeta_k$ and sum. With $\sum_k g_k \zeta_k / z_k = D_C / Z_C + O(\gamma)$, $\sum_k g_k \zeta_k E_C(z_k) = D_C E_C(Z_C) + O(\gamma^2)$, the term $\gamma \Lambda \sum_k g_k |\zeta_k|^2 = O(\gamma)$ and (I2),

$$D_C T_C = \frac{G_C^2 - S_C}{2} + O(\gamma). \quad (9)$$

Step 3: limits. Along the sequence, Λ_j is bounded, and so are $g_{k,j}$, $\zeta_{k,j}$ and $Z_{C,j}$; every subsequence therefore has a further subsequence along which the partition into clusters is fixed and $\Lambda_j \rightarrow \Lambda^*$, $g_{k,j} \rightarrow g_k^*$, $\zeta_{k,j} \rightarrow \zeta_k^*$ and $Z_{C,j} \rightarrow Z_C^*$, with $c \leq |g_k^*| \leq 1/c$, $c \leq |Z_C^*| \leq 1/c$, $|Z_C^* - Z_D^*| \geq c$ and $|\zeta_k^* - \zeta_l^*| \geq c$. We pass to such a subsequence. Since $\text{Im } \Lambda_j = 2\pi \text{Re } \kappa_j < 0$, $\text{Im } \Lambda^* \leq 0$. By Step 1, $z_{c,j} \rightarrow 0$ and $E_C(Z_C) \rightarrow 0$, so $T_{C,j} \rightarrow T_C^* = \Lambda^* Z_C^* - 1/Z_C^*$, and G_C , D_C , S_C converge to G_C^* , D_C^* and $S_C^* \geq c^2$. By (8) and (9),

$$G_C^* T_C^* = 0, \quad D_C^* T_C^* = \frac{G_C^{*2} - S_C^*}{2} \quad \text{for every cluster } C. \quad (10)$$

Step 4: the two branches. The stationary branch: some $T_C^* = 0$. Then $\Lambda^* \overline{Z_C^*} = 1/Z_C^*$, so $\Lambda^* = 1/|Z_C^*|^2$ is real and positive; since $\text{Re } \Lambda_j \rightarrow \Lambda^* > 0$ and $\text{Im } \Lambda_j \rightarrow 0$ with $\text{Im } \Lambda_j < 0$, (4) gives $P_j \rightarrow \infty$. In this branch the clusters C with $T_C^* = 0$ satisfy $G_C^{*2} = S_C^*$ by (10): they have their own harmonic condition in the limit, and a single vortex, for which $G_C^2 = S_C$ always, belongs here. The other clusters satisfy $G_C^* = 0$.

The translating branch: every $T_C^* \neq 0$. Then $G_C^* = 0$ for every C by (10), so every cluster has at least two vortices and vortices of both signs, and $D_C^* = -S_C^*/(2T_C^*) \neq 0$; in particular D_C does not tend to 0. (If $D_C^* = 0$ for some C , then (10) gives $G_C^{*2} = S_C^* \geq c^2$, so $G_C^* \neq 0$ and $T_C^* = 0$: the stationary branch.) For large j , $|T_{C,j}| \geq |T_C^*|/2$, so (8) gives $G_{C,j} = O(\gamma_j)$ and

$$\nu_{C,j} = \frac{G_{C,j}}{\gamma_j} = -\frac{\Lambda_j \overline{D_{C,j}} + D_{C,j}/Z_{C,j}^2}{T_{C,j}} + O(\gamma_j) \rightarrow \nu_C^* = -\frac{\Lambda^* \overline{D_C^*} + D_C^*/Z_C^{*2}}{T_C^*}.$$

Put $q_C^* = 1/(T_C^* Z_C^*) = p_C^* + iy_C^*$. Since $T_C^* Z_C^* = \Lambda^* |Z_C^*|^2 - 1$ and $D_C^* = -(S_C^*/2) q_C^* Z_C^*$,

$$\Lambda^* |Z_C^*|^2 = 1 + \frac{1}{q_C^*}, \quad q_{C,j} = -\frac{2D_{C,j}}{S_{C,j} Z_{C,j}} \rightarrow q_C^*, \quad (11)$$

and a direct computation gives

$$\nu_C^* = \frac{S_C^*}{2} (|q_C^*|^2 + \overline{q_C^*} + q_C^{*2}) = \frac{S_C^*}{2} (2p_C^{*2} + p_C^* + iy_C^*(2p_C^* - 1)). \quad (12)$$

The $\nu_{C,j}$ are real, so $y_C^*(2p_C^* - 1) = 0$ for every C . By (11), $\text{Im } \Lambda^* = -y_C^*/(|q_C^*|^2 |Z_C^*|^2)$ and $\text{Re } \Lambda^* = (1 + p_C^*/|q_C^*|^2)/|Z_C^*|^2$, for every C . The geometric meaning of q_C^* follows from (6) and (7): in the limit the conjugate velocity that the cluster induces on itself is $T_C^*/(2\pi i)$, and the one induced by the strong vortex is $1/(2\pi i Z_C^*)$, so their ratio is $V_C/V_s = \overline{T_C^* Z_C^*} = 1/\overline{q_C^*}$.

Case A: some $y_C^ \neq 0$.* Then $\text{Im } \Lambda^* \neq 0$, hence $\text{Im } \Lambda^* < 0$, and $y_D^* > 0$ for every cluster D . So $p_D^* = 1/2$ and $\nu_D^* = S_D^*/2$, that is, $\hat{\nu}_{D,j} = 2\nu_{D,j}/S_{D,j} \rightarrow 1$, for every D ; and $\text{Re } \Lambda^* > 0$. Since P is a continuous function of Λ on $\text{Im } \Lambda < 0$, P_j converges to

$$P^* = \frac{\text{Re } \Lambda^*}{-2 \text{Im } \Lambda^*} = \frac{|q_D^*|^2 + p_D^*}{2y_D^*} = \frac{y_D^{*2} + 3/4}{2y_D^*} = \frac{\sqrt{3}}{2} + \frac{(y_D^* - \sqrt{3}/2)^2}{2y_D^*} \geq \frac{\sqrt{3}}{2},$$

the same number for every D . Every y_D^* is therefore one of the two positive roots, y^* and $3/(4y^*)$, of $y^2 - 2P^*y + 3/4$, and by (11) clusters with different roots lie at different distances from the strong vortex. If $P^* = \sqrt{3}/2$, the roots coincide, every $y_D^* = \sqrt{3}/2$ and $q_D^* = e^{i\pi/3}$, and (11) gives the same $|Z_D^*|$ for every D .

Case B: every $y_C^ = 0$.* Then Λ^* is real. If $\Lambda^* \neq 0$, then $\text{Re } \Lambda_j$ is bounded away from 0 while $\text{Im } \Lambda_j \rightarrow 0$, so $P_j \rightarrow \infty$. If $\Lambda^* = 0$, then $1 + 1/p_C^* = 0$ by (11), so $q_C^* = -1$ and, by (12), $\nu_C^* = S_C^*/2$ for every C . Here the angular impulse enters. In the translating branch $\sum_i \Gamma_i z_i = \sum_C (\gamma G_C Z_C + \gamma^2 D_C) = O(\gamma^2)$, so $z_c = O(\gamma^2)$; the angular impulse $\sum_i \Gamma_i |z_i - z_c|^2 = \sum_i \Gamma_i |z_i|^2 - |z_c|^2 \sum_i \Gamma_i$ vanishes, and cluster C contributes $\gamma G_C |Z_C|^2 + 2\gamma^2 \text{Re}(\overline{Z_C} D_C) + O(\gamma^3)$ to $\sum_i \Gamma_i |z_i|^2$. Dividing by γ^2 and passing to the limit,

$$\sum_C (\nu_C^* |Z_C^*|^2 + 2 \text{Re}(\overline{Z_C^*} D_C^*)) = \sum_C \frac{S_C^*}{2} |Z_C^*|^2 (2p_C^{*2} - p_C^*) = 0, \quad (13)$$

where we used (12) and $D_C^* = -(S_C^*/2) q_C^* Z_C^*$. At $p_C^* = -1$ every term is $(3/2) S_C^* |Z_C^*|^2 > 0$, which is impossible. (In Case A, $p_C^* = 1/2$ and (13) holds term by term.)

Step 5: conclusion. Every subsequence has a further subsequence along which either $P_j \rightarrow \infty$ or Case A holds and $P_j \rightarrow P^* \geq \sqrt{3}/2$; this proves (S1). If $P_j \rightarrow \sqrt{3}/2$, only Case A with $P^* = \sqrt{3}/2$ can occur, so every subsequence has a further subsequence along which $q_{C,j} \rightarrow e^{i\pi/3}$ and $|Z_{C,j}|/|Z_{D,j}| \rightarrow 1$; these limits do not depend on the subsequence, so the whole sequence converges to them, which proves (S2). If $\sup_j P_j < \infty$, only Case A can occur, and the same argument gives $p_{C,j} \rightarrow 1/2$, $\hat{v}_{C,j} \rightarrow 1$ and $P_j - (y_{C,j}^2 + 3/4)/(2y_{C,j}) \rightarrow 0$; $\liminf_j y_{C,j} > 0$ because every subsequence has a further subsequence along which $y_{C,j}$ has a positive limit; for large j every cluster has vortices of both signs because $G_C^* = 0$ and $|g_k^*| \geq c$; and $\max_{k,l} |w_k - w_l| \rightarrow 0$ by (7). This proves (S3). The argument for Remark 1 is the same: a cluster of two or more vortices of one sign has $|G_C^*| \geq 2c$, so it can lie in neither branch, since the translating branch needs $G_C^* = 0$ and the stationary branch needs $G_C^{*2} = S_C^*$, while $G_C^{*2} - S_C^* \geq 2c^2$.

Proof of (d). Let k be the vortex of a single-vortex cluster, and let $\gamma \leq c^3/(20n)$, which is below γ_0 . Multiplying (6), where now $w_k = 0$, by z_k gives $\Lambda|z_k|^2(1 - \varepsilon_1) = 1 + \varepsilon_2$ with $\varepsilon_1 = \overline{z_c/z_k}$ and $\varepsilon_2 = z_k E_C(z_k)$. Now $|z_k| \geq c - \gamma/c \geq 19c/20$, the total circulation is at least $19/20$, and $|\sum_i \Gamma_i z_i| \leq n\gamma(1 + \gamma)/c^2$, so $|\varepsilon_1| \leq (20/19)^2(21/20)n\gamma/c^3 < 1.2n\gamma/c^3$. The other weak vortices are at distance at least $c - 2\gamma/c$ from z_k and $|z_k| \leq (1 + \gamma)/c$, so $|\varepsilon_2| \leq (n - 1)\gamma(1 + \gamma)/(c^3(1 - 2\gamma/c^2))$, which is at most $n\gamma/c^3$ because $\gamma \leq 1/(20n)$, $2\gamma/c^2 \leq 1/(10n)$ and $(n - 1)(20n + 1) \leq 2n(10n - 1)$. With $x = 10n\gamma/c^3 \leq 1/2$ we have $|\varepsilon_1| \leq 0.4x$ and $|\varepsilon_2| \leq 0.1x$, hence

$$|\Lambda|z_k|^2 - 1| \leq \frac{|\varepsilon_1| + |\varepsilon_2|}{1 - |\varepsilon_1|} \leq \frac{0.5x}{1 - 0.4x} \leq x,$$

because $x - 0.5x/(1 - 0.4x) = x(5 - 4x)/(2(5 - 2x)) \geq 0$. So $\text{Re}(\Lambda|z_k|^2) \geq 1 - x$ and $|\text{Im}(\Lambda|z_k|^2)| \leq x$, and (4) gives $P \geq (1 - x)/(2x) \geq 1/(4x) = c^3/(40n\gamma)$.

Sharpness. The class of [6, Theorem 3] lies in $\mathcal{K}(2m, c^2)$, and, as shown after the proof of [6, Theorem 3], the collapses of three vortices and of two rings with a central vortex [6, Corollary 1, Proposition 3] lie in that class, with $m = 1$ and $c = 1/2$ and with $m \geq 2$ and $c = \min(1/2, 2\sin(\pi/m))$, and have $P \rightarrow \sqrt{3}/2$. So $\sqrt{3}/2$ in (a) cannot be increased for $n = 2m$. $\square$

For $n = 1$, part (d) gives $P \rightarrow \infty$, so (a) holds with any constant. For odd $n \geq 3$ we have only numerical evidence that $\sqrt{3}/2$ is sharp: a strong vortex with one triple ($n = 3$) or with a pair and a triple ($n = 5$) has least P tending to $\sqrt{3}/2$, from below and from above respectively (Table 1).

3.3 Beyond leading order: why four vortices go below $\sqrt{3}/2$

The following expansion is formal: we carry the equations of Step 2 one order further in γ without controlling the remainders, and we confirm the result numerically. For a cluster with $G_C = O(\gamma)$ let $M_C = \sum_k g_k \zeta_k^2$, $N_C = \sum_k g_k |\zeta_k|^2$ and

$$J_C = N_C - \text{Re}\left(\overline{M_C} \frac{D_C}{\overline{D_C}}\right) = 2 \sum_{k \in C} g_k h_k^2, \quad e_C = \frac{\gamma J_C S_C}{2|D_C|^2},$$

where h_k is the coordinate of ζ_k perpendicular to the dipole D_C ; J_C does not depend on the point Z_C from which ζ_k is measured. Measured from the point where $M_C = 0$, the next term of (9) is $D_C T_C = -S_C/2 - \gamma \Lambda N_C + O(\gamma^2)$, and with $L = \Lambda|Z_C|^2$ this becomes $L = 1 + (1 + e_C L)/q_C$, so $L = (q_C + 1)/(q_C - e_C)$. Carrying (8) to the same order gives $\nu_C/(S_C/2) = (|q_C|^2 + \overline{q_C} + q_C^2 - e_C q_C)/(1 + e_C)$, which is real only if $y_C(2p_C - 1 - e_C) = 0$. On $p_C = (1 + e_C)/2$,

$$P = \frac{(3 + e_C)(1 - e_C)/4 + y_C^2}{2y_C(1 + e_C)} \geq \frac{\sqrt{(3 + e_C)(1 - e_C)}}{2(1 + e_C)} = \frac{\sqrt{3}}{2} - \frac{2}{\sqrt{3}} e_C + O(e_C^2).$$

clusters beside a strong vortex +1	$\min_C e_C/\gamma$	slope of (14)	$(P_{\min} - \sqrt{3}/2)/\gamma$ at $\gamma =$		
			10^{-2}	10^{-3}	10^{-4}
pair (0.9, -0.9)	0	0	0.010428	0.001051	0.000105
triple (1, 0.6, -1.6)	0.73469	-0.84835	-0.79999	-0.84347	-0.84786
triple (0.8, -1.3, 0.5)	0.60465	-0.69819	-0.67852	-0.69624	-0.69800
quadruple (1, 0.7, -0.5, -1.2)	0.15302	-0.17670	-0.11027	-0.17009	-0.17604
quadruple (1, 0.9, 0.8, -2.7)	1.83673	-2.12088	-1.99482	-2.10832	-2.11962
pair and triple (1, 0.6, -1.6)	0	0	0.010580	0.001002	0.000098
two triples and a quadruple	0.15302	-0.17670	-0.11815	-0.17077	-0.17611
triple (-1, -0.6, 1.6)	-0.73469	0.84835	0.89766	0.85324	0.84884

Table 1: The formal expansion (14) against exact collapses computed at fifty digits (`verify_cluster_collapses.py`; numerical). Cluster circulations are the g_k of the cluster; $P_{\min}$ is minimized over y with the cluster shapes and directions fixed. The error of the slope falls by a factor between 9.9 and 10.6 per decade in every row. With a pair present, $(P_{\min} - \sqrt{3}/2)/\gamma^2$ is 1.043, 1.051, 1.052 (pair) and 1.058, 1.002, 0.983 (pair and triple).

All clusters share Λ , so each gives such a lower bound, and the least P over a family in which the cluster shapes and the directions of the clusters are fixed is, formally,

$$P_{\min} = \frac{\sqrt{3}}{2} - \frac{2}{\sqrt{3}} \min_C e_C + O(\gamma^2). \quad (14)$$

For a pair both vortices lie on the axis of the dipole, so $J_C = 0$ and $e_C = 0$: there is no term of order γ , in agreement with [6, Proposition 4], where the first correction is of order γ^2 and positive. For a triple with $G_C = 0$ the translating relative equilibria we use are equilateral, and then $e_C = -3\gamma g_1 g_2 g_3 / S_C$, so the slope in (14) is $2\sqrt{3} g_1 g_2 g_3 / S_C$. With a positive strong vortex this is negative for a triple of signs $(+, +, -)$: such a triple goes below $\sqrt{3}/2$ at first order in γ , while a triple $(+, -, -)$ stays above. For the triple $g = (1, 0.6, -1.6)$ the slope is $2\sqrt{3}(-0.96)/3.92 = -0.84835\dots$; this number belongs to that triple only.

Table 1 compares (14) with exact collapses. For each arrangement the cluster shapes are translating relative equilibria of zero net circulation, the circulations of each cluster except one and the directions of the cluster centers are fixed, and the remaining unknowns (all positions, Λ and one circulation per cluster) are found by Newton's method at fifty digits along the one-parameter family parametrized by y of the binding cluster; $P_{\min}$ is the least P along that family, so it is minimized over y only, with the shapes and directions fixed. Every point used is accepted only after the Biot-Savart velocities (2) of all vortices, the strong one included, agree with $\kappa(z_j - z_c)$ to 10^{-40} relative. In every arrangement the error of the slope in (14) shrinks by a factor between 9.9 and 10.6 per decade of γ , which is the $O(\gamma^2)$ remainder; with a pair in the arrangement the pair binds, $\min_C e_C = 0$, and $P_{\min} > \sqrt{3}/2$.

This explains the structure found in [6]. Three vortices are a strong vortex and a pair, $e = 0$, and $P > \sqrt{3}/2$ [6, Corollary 1]. The first configuration with a term of order γ is a strong vortex with a triple, four vortices, and with signs $(+, +, -)$ that term is negative. The certified four-vortex minimum $P_4 = 0.7978967838\dots$ of [6, Theorem 4] has exactly this structure: one strong vortex and a $(+, +, -)$ triple with circulations 0.104, 0.167 and -0.227 times the strong one, whose net circulation is 0.978 times $(1/2) \sum_k \Gamma_k^2$ (in units of the strong circulation), as Theorem 1(b) predicts for weak clusters. There the triple is not weak (its diameter is 0.23 times its distance from the strong vortex), so (14) is only a qualitative guide: it predicts the sign of the effect, not the value.

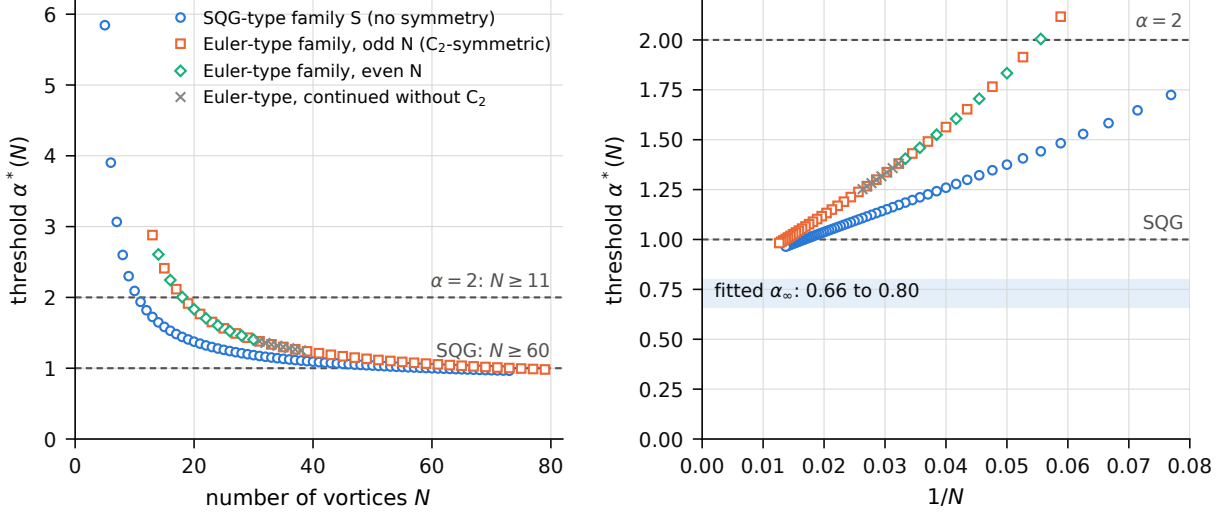


Figure 1: The least α at which N vortices collapse self-similarly without rotation, along four families (numerical). Left: against N ; the dashed lines are $\alpha = 2$ and SQG ($\alpha = 1$), crossed by the SQG-type family at $N = 11$ and $N = 60$. Right: against $1/N$; the band is the range 0.66 to 0.80 of the limits α_∞ given by the fits of Table 3. The data are in `data/phase-diagram/alpha-star-phase-diagram.csv`.

4 The phase diagram of collapse without rotation

Everything in this section is numerical (binary64). Let Z be the set of self-similar collapses without rotation of N vortices, with α free, in the gauge $2\pi\kappa = -1$ with the collision point at the origin; near a point where the equations are independent it is a manifold of dimension $N - 1$. Following a family of numerical minimizers of P as α increases, its least P first reaches 0 at a threshold $\alpha^*(N)$; at such a point α has a local minimum on Z . We locate the thresholds by continuation in α to within 10^{-4} and then solve the first-order (Karush–Kuhn–Tucker) system of the minimization of α on Z , which is square, by Newton’s method; the reduced Hessian of the Lagrangian on the tangent space of Z is positive definite at every threshold below, so each is a strict local minimum of α on Z (numerically). The refined values lie inside the recorded brackets, and the program `verify_phase_diagram.py` rechecks every one from its stored configuration (Section 7).

4.1 Two families

The *SQG-type family S* has one strong vortex near the collision point, one strong vortex of the opposite sign, and weak vortices of one sign, with no symmetry; the least minimizers of P at $\alpha = 0$, 1 and 2 for $N = 8$, 9 and 10 all continue in α to its thresholds. We computed it for $5 \leq N \leq 73$, the new members for $N \geq 60$ by inserting a passive vortex at a stagnation point of the similarity flow and minimizing α again. The *Euler-type family E* comes from the Euler minimizers of [6, Sect. 7]; its odd- N thresholds are exactly C_2 -symmetric (a central vortex and pairs $\pm w$ with equal circulations; the defect is 10^{-13}), while the even ones are not. We extended the odd branch to $N = 79$ with the symmetric system reduced to the fixed-point set of the symmetry, and at all 34 odd thresholds from $N = 13$ to 79 the reduced Hessian in the full space is positive definite, so the symmetric points are strict local minima of α on Z without the symmetry imposed. A continuation

α	6	5	4	3	2.5	2	1.5	1.2	1
least N	5	6	6	8	9	11	17	29	60

Table 2: The least number of vortices with a self-similar collapse without rotation, along the families of Figure 1 (numerical; all from the family S).

of E from $N = 30$ without the symmetry gives eight more points ($31 \leq N \leq 38$), slightly below the odd branch at odd N .

The thresholds are shown in Figure 1. Along both families $\alpha^*(N)$ decreases strictly with N , and S is the lowest family at every N where another was computed. The least N with a collapse without rotation, along these families, is given in Table 2. For $\alpha = 2$ the values are $\alpha^*(9) = 2.3000208434$, $\alpha^*(10) = 2.0913751011$ and $\alpha^*(11) = 1.9372878365$, so eleven is the least number along S , in agreement with the certified collapse of [6, Theorem 5(a)]. For SQG, S crosses $\alpha = 1$ between $N = 59$ (1.00246) and $N = 60$ (0.99921), in agreement with [6, Theorem 5(b)], and the odd branch of E crosses it between $N = 73$ (1.00246) and $N = 75$ (0.99543), so it also gives SQG collapses without rotation, C_2 -symmetric ones, from $N = 75$. For $N > 73$ only that branch was computed. The least values computed are 0.96443 (S , $N = 73$) and 0.98242 (E , $N = 79$).

4.2 The searches at $\alpha = 2$ and at four vortices

Whether ten vortices can collapse without rotation at $\alpha = 2$ is not settled by one family, so we searched further. Continuation in α of all eighteen recorded local minima of P for $N = 8, 9$ and 10 at $\alpha = 0, 1$ and 2 up to $P = 0$, followed by the minimization of α on Z , ends at the thresholds of S or at larger ones (2.5993690531, 5.7554694090, 5.2557582423, 4.4051046186 and 8.0863308796). A random walk on Z with large perturbations of positions, circulations and signs, each step followed by the minimization of α , made 5352 attempts with 323 landings for $N = 9$ and 5221 attempts with 277 landings for $N = 10$, and found no local minimum of α below 2.3000208434 and 2.0913751011. A perturbation search for the least P at $\alpha = 2$ (1110 attempts and 358 landings for $N = 9$, 652 and 191 for $N = 10$) found the least values 0.0675758057 and 0.0237078013, and no point with $P = 0$. Driving the circulation of one vortex of the $\alpha = 2$, $N = 11$ collapse to zero while minimizing α ends, for each of the nine weak vortices, at 2.0913751011, the ten-vortex threshold of S ; for the strong vortex of opposite sign the homotopy fails. Reduced C_2 -symmetric systems give no lower thresholds: the least values found are 5.7555 ($N = 9$, a center and four pairs), 4.9340 ($N = 10$, five pairs), 3.7369 ($N = 11$), 3.3600 ($N = 12$) and 2.8803 ($N = 13$). This is evidence, not a proof, that ten vortices cannot collapse without rotation at $\alpha = 2$.

For four vortices no clean threshold exists along the families we followed: the $\alpha = 2$ four-vortex minimum continues upward, $P = 0.207$ at $\alpha = 5$ and 0.069 at $\alpha = 10$, and reaches $P = 0$ near $\alpha = 14.63$, but there three circulations are 10^{-6} to $6.5 \cdot 10^{-6}$ times the fourth: a strong vortex with near-passive companions. Three vortices cannot collapse without rotation for any $\alpha > -2$ [6, Theorem 2].

4.3 The limit of many vortices

For $20 \leq N \leq 50$ the decrements of S are close to $11/N^2$ ($N^2|\alpha^*(N+1) - \alpha^*(N)|$ lies between 11.03 and 11.8), which suggests $\alpha^*(N) \approx \alpha_\infty + 11/N$. This law does not hold: $N^2|\alpha^*(N+1) - \alpha^*(N)|$ has its least value 11.03 at $N = 35$ and rises to 11.9 at $N = 72$, and the local exponent p of $\alpha^* \approx a + bN^{-p}$ falls steadily, from 1.06 at $N = 30$ to 0.90 at $N = 48$ and 0.82 at $N = 66$, so the approach is slower than $1/N$. Along the odd branch of E the same quantity stops falling at 19.24

family	form	$N \geq 56$ (S), $N \geq 63$ (E)		$N \geq 33$ (S), $N \geq 40$ (E)	
		α_∞	rms	α_∞	rms
S	$a + b/N + c/N^2$	0.789	$5.4 \cdot 10^{-6}$	0.800	$1.5 \cdot 10^{-4}$
S	$a + b/N + c/N^2 + d/N^3$	0.776	$2.3 \cdot 10^{-7}$	0.785	$1.8 \cdot 10^{-5}$
S	$a + bN^{-p}$	0.767	$2.9 \cdot 10^{-6}$	0.787	$1.1 \cdot 10^{-4}$
S	$a + b/(\ln N)^q$	0.677	$4.0 \cdot 10^{-7}$	0.684	$2.1 \cdot 10^{-5}$
S	$b/(\ln N)^q$, $\alpha_\infty = 0$ forced	0	$2.1 \cdot 10^{-4}$	0	$2.4 \cdot 10^{-3}$
E	$a + b/N + c/N^2 + d/N^3$	0.724	$1.4 \cdot 10^{-7}$	0.731	$8.9 \cdot 10^{-6}$
E	$a + b/(\ln N)^q$	0.656	$1.4 \cdot 10^{-6}$	0.670	$5.2 \cdot 10^{-5}$
E	$b/(\ln N)^q$, $\alpha_\infty = 0$ forced	0	$3.2 \cdot 10^{-4}$	0	$3.2 \cdot 10^{-3}$

Table 3: Fits of the thresholds of Figure 1 over trailing windows (numerical; a selection; all forms and a middle window are in `data/phase-diagram/alpha-inf-fits.json`).

near $N = 69$ and turns up. Table 3 fits five forms to the tails of both families, over three windows each. Every fit with α_∞ free gives a positive limit, between 0.677 and 0.800 for S and between 0.656 and 0.766 for E : the form $a + b/(\ln N)^q$, which follows the drifting exponent, gives 0.66 to 0.68, the polynomial and power forms 0.72 to 0.80. Fits that force $\alpha_\infty = 0$ have a root-mean-square residual 30 to 530 times that of the same form with α_∞ free, and 130 to 3400 times that of the best free fit in the window.

This is consistent with no Euler collapse without rotation along these families. It says nothing about other families, and an approach slower than any power of $\ln N$ is not excluded by $N \leq 79$. Since the first-order system of each threshold is square and nondegenerate, the interval Newton (Krawczyk) method used in [6, Sect. 8] could certify any $\alpha^*(N)$ rigorously; we have not done so.

5 Structure and stability of the collapses without rotation

5.1 No mirror symmetry

The following observation is elementary; we record it because it rules out a natural ansatz for collapses, with or without rotation.

Proposition 1. *Let α be real and consider a self-similar collapse of $N \geq 2$ point vortices under (1) in which all circulations and their sum are nonzero. No reflection of the plane maps the configuration onto itself, either with every circulation kept or with every circulation reversed.*

Proof. Suppose a reflection σ maps the configuration onto itself, with a permutation π such that $\sigma(z_j) = z_{\pi(j)}$. If every circulation is reversed, $\Gamma_{\pi(j)} = -\Gamma_j$, then $\sum_j \Gamma_j = \sum_j \Gamma_{\pi(j)} = -\sum_j \Gamma_j$, so the total circulation vanishes, contrary to the hypothesis. Suppose every circulation is kept, $\Gamma_{\pi(j)} = \Gamma_j$. Then σ fixes the center of vorticity, so after a translation and a rotation, under which (1) is equivariant and κ does not change, $z_c = 0$ and $\sigma(z) = \bar{z}$. The right side of (1) satisfies $V_j(\bar{z}_1, \dots, \bar{z}_N) = -\overline{V_j(z_1, \dots, z_N)}$ for real circulations: the mirror image of a motion, run backwards, is a motion. Hence $V_{\pi(j)} = -\overline{V_j}$. The collapse gives $V_j = \kappa z_j$ and $V_{\pi(j)} = \kappa z_{\pi(j)} = \kappa \bar{z}_j$, so $\kappa \bar{z}_j = -\bar{\kappa} \bar{z}_j$ for every j , and since some $z_j \neq 0$, $\text{Re } \kappa = 0$, contrary to $\text{Re } \kappa < 0$. $\square$

Rotational symmetry is allowed: the odd branch of E in Section 4 is C_2 -symmetric. By Proposition 1 no member of the families of [6, Theorem 5] is mirror symmetric; `verify_family_structure.py` checks the time-reversal identity on random configurations for $\alpha = 0, 0.5, 1$ and 2 , and that the mirror image of the $\alpha = 2$ collapse expands with $2\pi\kappa' = 1$.

5.2 The nine-parameter family at $\alpha = 2$

The rest of this section is numerical (binary64). At the eleven-vortex collapse of [6, Theorem 5(a)], with the rotation and the scale fixed and $2\pi\kappa = -1$, the 22 equations have full rank in 31 unknowns (the ratio of the smallest to the largest singular value is $7.3 \cdot 10^{-4}$), so the collapses without rotation form a manifold of dimension nine there. We continued along both signs of each of the nine tangent directions for 300 steps, to arclengths 1.9 to 3.9 relative to the size of the configuration: no collision (the closest pair stays above 0.045 of the size), no vanishing circulation and no loss of rank occurs, and the sign pattern, one strong vortex, one of the opposite sign and nine weak vortices of the sign of the strong one, never changes, while the ratio of the largest to the smallest circulation varies from 17 to 153.

The family has no C_5 -symmetric member. Eleven vortices with C_5 symmetry form two regular pentagons and a central vortex; for two n -gons at radii 1 and ρ , relative rotation θ and a central vortex, equal radial rates force the circulation $-1/\rho^2$ on the second ring, the central circulation enters linearly, and collapse without rotation reduces to one scalar equation $F(\rho, \theta) = 0$. On a 200×200 grid, $0.05 \leq \rho \leq 20$, $0 < \theta < 2\pi/n$, F has one sign for $n = 2, 3$ and 5 at $\alpha = 1$ and 2, with $|F|$ at least 0.03 times its natural scale, so no collapse without rotation of this form exists for five, seven or eleven vortices at these α . C_2 -symmetric collapses without rotation of eleven vortices (a center and five pairs) exist, but the least α found for them is $3.7369117731 > 2$ (Section 4.2). We found no simplest member: minimizing the spread of the nine weak circulations along the family stalls with circulations that still differ by a factor 1.32.

5.3 Linear stability

In similarity variables $z = \rho(t)\zeta$, $d\tau = \rho^{-2\beta}dt$ with $\beta = 1 + \alpha/2$, a self-similar collapse is a fixed point of $d\zeta/d\tau = V(\zeta) - \kappa\zeta$. With the circulations held fixed and $2\pi\kappa = -1$, a mode of the linearization with exponent k (the real part of an eigenvalue of the Jacobian of $2\pi V(\zeta) + \zeta$) grows relative to the size r of the configuration like r^{-k} . The Jacobian of $2\pi V$ is a Hamiltonian matrix with respect to the symplectic form weighted by the circulations, so its eigenvalues come in pairs $\pm\mu$ and the exponents $k = 1 + \text{Re } \mu$ in pairs with $k + k' = 2$, and six of them, 0, 2β , 1, 1, 2 and $-\alpha$, belong to the symmetries; both facts hold to 10^{-12} and $4 \cdot 10^{-11}$ in the computation and check it. For the eleven vortices at $\alpha = 2$, 8 of the 16 shape modes are unstable, all real, with k from 8.43 to 258.15; for the sixty SQG vortices, 57 of the 114 shape modes are unstable, all real, with k from 5.04 to 4273.65.

Table 4 shows direct integrations (the DOP853 method with relative tolerance 10^{-13} , which conserves the Hamiltonian to $3 \cdot 10^{-10}$, the impulse to 10^{-14} and the angular impulse to 10^{-13}). A random perturbation of relative size 10^{-6} grown at the largest exponent alone reaches 10^{-2} at the size $10^{-4/k_{\max}}$, that is 0.965 ($\alpha = 2$) and 0.998 (SQG); the perturbed motions depart from the self-similar collapse, by a shape deviation of 10^{-2} , at sizes 0.952 to 0.956 and 0.991 to 0.992, having turned by less than 10^{-4} rad. After departure the configuration deforms and turns (by up to 0.34 and 0.23 rad in the window), never gets below 0.828 ($\alpha = 2$) and 0.887 (SQG) of its initial size, and no close pair forms. Even unperturbed, the binary64 integrations depart at sizes 0.882 and 0.983 from round-off alone, so a direct simulation cannot exhibit these collapses. Lewkowicz and Kudela found the same sensitivity for a seven-vortex Euler collapse, where the scale reached before departure grows with the working precision [11], without computing exponents.

collapse	perturbation	departure		rotation before departure (rad)	least size (t/t_c)	closest pair / R_0
		size	t/t_c			
$\alpha = 2, N = 11$	none	0.8819	0.395	$4 \cdot 10^{-5}$	0.796 (0.99)	0.075
	10^{-6} , seed 1	0.9524	0.177	$4 \cdot 10^{-5}$	0.851 (0.59)	0.085
	10^{-6} , seed 2	0.9529	0.175	$2 \cdot 10^{-6}$	0.828 (0.61)	0.092
	10^{-6} , seed 3	0.9558	0.165	$2 \cdot 10^{-5}$	0.859 (0.55)	0.095
SQG, $N = 60$	none	0.9834	0.049	$8 \cdot 10^{-5}$	0.895 (0.41)	0.0064
	10^{-6} , seed 1	0.9908	0.027	$7 \cdot 10^{-5}$	0.901 (0.47)	0.0061
	10^{-6} , seed 2	0.9915	0.025	$7 \cdot 10^{-5}$	0.887 (1.20)	0.0059
	10^{-6} , seed 3	0.9915	0.025	$6 \cdot 10^{-5}$	0.904 (0.50)	0.0050

Table 4: Direct integration of the two collapses without rotation of [6, Theorem 5] to $1.5 t_c$ ($\alpha = 2$) and $1.2 t_c$ (SQG), with and without a random perturbation of the positions of relative size 10^{-6} (`verify_stability.py`; numerical). Sizes are relative to the initial size R_0 ; the SQG run with seed 2 is at its least size when the window ends.

6 The continuum limit

Everything in this section is numerical, in high-precision arithmetic, and we make no claim of priority for it. O’Neil [13] reports numerical evidence for collapse configurations of vortex sheets, computed with point vortices that approximate the sheets; we have seen only its abstract and Kudela’s description of it [9, p. 2], according to which the sheets there are accompanied by a strong vortex, their shape is prescribed first and then replaced by identical weak point vortices, and the resulting algebraic system, a point-vortex approximation of the Birkhoff–Rott equation, is solved together with an invariant of the motion. Kudela [9] observed that in numerical self-similar collapses of fifty vortices with one, two or four strong vortices the weak vortices line up on arcs, which he calls vortex sheets. We have not been able to compare these configurations with ours.

In the two-arm family of Euler minimizers of [6, Sect. 7], a vortex at the center and two arms of weak positive vortices carry two negative vortices, and P decreases with N towards about 0.47736. As N grows, the weak vortices of each arm approach a curve with a smooth circulation density. We model the limit by a point-symmetric vortex sheet $z(x)$, $-1 \leq x \leq 1$, with $z(-x) = -z(x)$, $|dz/dx| = L$ constant (so x is proportional to arclength) and the tip $z(1) = 1$, carrying the circulation $g(x) dx$ with $g(x) = \sqrt{1 - x^2} h(x)$, together with two point vortices of circulation -1 at $\pm z_n$. Self-similar collapse requires

$$\text{PV} \int_{-1}^1 \frac{g(t) dt}{z(x) - z(t)} - \frac{1}{z(x) - z_n} - \frac{1}{z(x) + z_n} = \Lambda \overline{z(x)} \quad (|x| < 1), \quad \int_{-1}^1 \frac{g(t) dt}{z_n - z(t)} - \frac{1}{2z_n} = \Lambda \overline{z_n}, \quad (15)$$

the principal-value Birkhoff–Rott integral on the sheet and the velocity at the point vortex, with Λ as in (4) and $P = \text{Re } \Lambda / (-2 \text{Im } \Lambda)$. The continuum form of $\sum_{i < j} \Gamma_i \Gamma_j = 0$, with the sheet of total circulation Σ contributing $\Sigma^2/2$ from its own pairs, is $1 - 2\Sigma + \Sigma^2/2 = 0$, whose roots are $2 \pm \sqrt{2}$.

We discretize the tangent angle of the sheet and h as even Chebyshev series of m terms each, evaluate the principal-value integral by Gauss–Chebyshev quadrature of the second kind with n nodes, collocate at the positive zeros of the Chebyshev polynomial T_{n+1} , and solve the overdetermined collocation system in the least-squares sense, so that its residual measures the truncation error. The solutions form a one-parameter family, and we minimize P along it. The solutions were refined in ball arithmetic (Arb, through python-flint) at 192 and 256 bits, from a binary64 Jacobian and Arb residuals. They converge exponentially in m (Table 5): the residual falls from $3.7 \cdot 10^{-18}$ at $m = 64$

m	n	bits	P	collocation residual
64	384	192	0.47736353369161202484304865...	$3.7 \cdot 10^{-18}$
80	480	192	0.4773635336916120248430486309556...	$1.7 \cdot 10^{-22}$
100	800	256	0.47736353369161202484304863090601204...	$7.0 \cdot 10^{-28}$
120	960	256	0.4773635336916120248430486309060121246264...	$2.5 \cdot 10^{-33}$
140	840	256	0.4773635336916120248430486309060121246263...	$7.6 \cdot 10^{-39}$

Table 5: The least winding of the continuum model (15) at five resolutions (numerical; m Chebyshev terms per unknown function, n quadrature nodes; `verify_continuum_limit.py`).

to $7.6 \cdot 10^{-39}$ at $m = 140$, the values of P at $m = 100$ and $m = 120$ differ from that at $m = 140$ by $8.3 \cdot 10^{-35}$ and $1.3 \cdot 10^{-40}$, a factor $6 \cdot 10^5$ per twenty terms, and so, if that rate persists,

$$P_\infty = 0.4773635336916120248430486309060121246263 \dots$$

to forty digits. At the solution the sheet carries $2 + \sqrt{2}$, each arm $1 + 1/\sqrt{2}$, and the angular impulse vanishes, to 10^{-43} . Independently of the collocation, the continuous equations (15) hold between the collocation points to $2 \cdot 10^{-13}$, with the principal value computed by subtracting the singularity and adaptive quadrature at forty digits from the stored coefficients rounded to binary64. Along the family, $dP/dc = 0$ and $d^2P/dc^2 = 0.1096 > 0$ in the family coordinate c , so P_∞ is a local minimum along it; we restricted the density to vanish like a square root at the tips, and whether another tip behavior gives a smaller P is not settled here.

The limit agrees with the finite family: a fit $P(N) = a + b/N + \dots + e/N^4$ to the 31 members with $N \geq 301$ of [6, Sect. 7], whose values are stored to thirteen digits, gives $a = P_\infty - 9 \cdot 10^{-13}$, and fixing $a = P_\infty$ leaves the residual at the rounding level, $2.7 \cdot 10^{-14}$; cubic Richardson extrapolation in $1/N$ through $N = 101, 203, 403$ and 603 gives 0.477363529 . In the limit the sheet has half-length $L = 1.0306552905$, the point vortices sit at $\pm(0.7678760388 + 0.1405296657i)$ and $\Lambda = 4.2243187321 - 4.4246349312i$; the circulation per unit length is 1.3190 at the center and largest, 2.3308 , at 0.707 of the half-length, and it vanishes like a square root at the tips; the tangent of the sheet turns from 9.32° at the center to 33.98° at the tip. We have found no closed form for P_∞ : with forty digits, no algebraic relation of degree at most five with integer coefficients up to 10^6 holds for P_∞ , P_∞^2 , $1/P_\infty$, $\arctan(2P_\infty)/\pi$ or $4P_\infty^2 + 1$, and no integer relation with coefficients up to 10^6 connects P_∞ , P_∞^2 or $1/P_\infty$ with 1 and three or four of $\sqrt{2}$, $\sqrt{3}$, π , $\ln 2$, $\ln 3$, e and $\ln(1 + \sqrt{2})$; at these sizes a relation found by chance would need larger coefficients.

7 Numerical verification

Table 6 lists the programs. Each prints every check with its result, stops with an error if one fails, and writes its report to `data/`; under Python 3.11 with the package versions of `code/requirements.txt` each runs in about two minutes or less, the stability program in its full form in two and a half. The programs `verify_phase_diagram.py`, `verify_family_structure.py` and `verify_stability.py` share the module `collapse_core.py` (the law (1), its analytic Jacobian, Newton's method and the first-order system of the minimization of α), and `verify_continuum_limit.py` uses `continuum_model.py` (the discretization, in binary64) and `continuum_arb.py` (the same residual in Arb, and the refinement, which also runs as a program). The convergence checks of `verify_cluster_collapses.py` require the error of each asymptotic statement to fall by a factor of at least five per decade of γ , instead of fixed tolerances. The inputs taken from [6], the two certified collapses without rotation, the four-vortex minimum and the two-arm family, are copied into `data/`.

Table 6: The verification programs, what they check, and the number of checks.

Program	What it checks	Checks
<code>verify_cluster_identities.py</code>	Every identity in the proof of Theorem 1 and of Section 3.3, exactly (SymPy): (I1), (I2), the expansions behind (8) and (9), (11), (12), (13), the constants of (d) for every n , Remark 3, the formal expansion and the triple slope; negative controls	62
<code>verify_cluster_remainders.py</code>	The remainders of (8), (9) and (7) on 150 random configurations of $\mathcal{K}(n, 0.3)$, $\gamma = 10^{-2}$ to 10^{-7} , bounded after division by their order, and a control that grows like $1/\gamma$; the constants of Step 1 and of (d) on 400 random configurations with up to 40 weak vortices	8
<code>verify_cluster_collapses.py</code>	Exact collapses at fifty digits for eight cluster arrangements at $\gamma = 10^{-2}, 10^{-3}, 10^{-4}$ (Table 1), with convergence tests of (14) and of part (b); singletons and (d) on rings with a strong center; the controls of Remark 1 and Section 3.3. About two minutes (<code>--quick</code> : one minute, 72 checks)	90
<code>verify_cluster_stored.py</code>	The 21 stored collapses rechecked with separate code at sixty digits: least-squares κ over all vortices, both invariants, P , membership in $\mathcal{K}(n, c)$ with $c \geq 0.37$, and the exact net circulation of Remark 3	26
<code>verify_phase_diagram.py</code>	All 120 thresholds of Figure 1 from their stored configurations: collapse without rotation, first-order conditions, reduced Hessians, C_2 defects; Tables 2 and 3; the $11/N$ law and the local exponent; the end points of the searches of Section 4.2	46
<code>plot_phase_diagram.py</code>	Figure 1 from the table	
<code>verify_family_structure.py</code>	Section 5.2: the dimension nine, 18 continuations, the time-reversal identity of Proposition 1 and the mirror image, the ring reduction against Biot–Savart and its one-signed F , the most nearly equal weak circulations	15
<code>verify_stability.py</code>	Section 5.3: symmetry exponents, the pairing $k + k' = 2$, the unstable modes, and the eight integrations of Table 4. Two and a half minutes (<code>--quick</code> : SQG integrations to $0.08 t_c$, 20 s, 13 checks)	14
<code>verify_continuum_limit.py</code>	Section 6: convergence in m , the two finest solutions recomputed in Arb, $\Sigma = 2 + \sqrt{2}$ (SymPy), the residual between collocation points, local minimality, the finite family, the shape, and the closed-form searches	13

Program	What it checks	Checks
<code>continuum_arb.py</code>	Run as a program: the refinement of Section 6 at $m = 64$, $n = 384$ from sixteen digits; it reproduces the stored value to $9 \cdot 10^{-25}$, well inside that resolution's residual $3.7 \cdot 10^{-18}$	

8 Discussion

Theorem 1 extends [6, Theorem 3] from pairs to clusters, and its proof follows the same compactness scheme; the new ingredients are the identities (I1) and (I2) of the internal field, which reduce every cluster at leading order to its net circulation G_C and its dipole D_C , and the observation that the leading-order equations of a cluster are those of a pair with u_j replaced by q_C . For one pair, Krishnamurthy and Stremler describe a small dipole in the field of a strong vortex [8, Sect. 3.6, Fig. 9], as the companion paper records. That a vortex pair with separation ε and net circulation of order ε^2 behaves as a charged particle is the scaling of Drivas, Glukhovskiy and Khesin [4, Theorem 1.1], who prove it for pairs on surfaces without a strong vortex and without collapse; our clusters have the same scaling, net circulation of order γ^2 at size γ . Leoncini, El Kettani and Ugalde [10, Sect. B] expand in the size of a small triple whose net circulation equals that of a parent vortex, the opposite case of ours, and do not discuss winding. Lydon, Nazarenko and Laurie [12] study dipoles scattering on clusters of like-signed vortices. In the literature we have read we have found no treatment of near-neutral clusters in a collapse, and no bound on the winding for them; Kudela's weak vortices [9] are like-signed and spread along arcs, not near-neutral clusters.

For collapses without rotation with more than three vortices, the only results we know are those of [6]. The α -model collapse literature we have read treats three vortices, or, as Grotto and Pappalettera [5] do, collapses and bursts of many vortices constructed from a three-vortex self-similar one, which are not self-similar; Badin and Barry [2] derive the conditions for collapse in these models. We have not found a map of the thresholds $\alpha^*(N)$ in the literature we have read; ours is numerical and covers two families only.

The linear stability of self-similar collapse is classical for three vortices; see the discussion and references in [6, Sect. 9]. For more vortices, Zbarsky [14, p. 20] notes that stability of some self-similarly expanding system of more than three point vortices would extend his Corollary 16 to that system, and Lydon, Nazarenko and Laurie [12, p. 39] raise the stability of vortex collapse as a question; Lewkowicz and Kudela [11] observed the sensitivity numerically. Section 5.3 computes the exponents for the two collapses without rotation. By Proposition 1 the mirror image of a collapse is an expansion with the same shape, whose exponents are those of the collapse with the sign of time reversed.

The continuum model of Section 6 is a vortex sheet with two point vortices that shrinks self-similarly onto a point, unlike self-similar spiral sheets that roll up and grow; the collapsing sheets of O'Neil [13] and the sheet-like arcs of Kudela [9] are the nearest objects we know of, and until O'Neil's paper has been read we do not claim that the continuum model or its minimum is new.

Meaning and limits. A strong vortex that gathers weak vortices into neutral-looking clusters still turns, while shrinking by the factor 10, by nearly $(\sqrt{3}/2) \ln 100 \approx 4$ radians if the clusters are weak, whatever their number and shape; to turn less, the weak vortices must be strong enough, or organized differently, and the first way to do it, a $(+, +, -)$ triple, is the structure of the certified four-vortex minimum. Collapse without rotation needs much more: along our families it needs $\alpha > 0.96$ for up to 79 vortices, and the thresholds do not appear to tend to the Euler value. The

collapses without rotation that exist are unstable in many directions, so they are exceptional motions, not attractors. All results concern point vortices; finite cores regularize the collision.

9 Open problems

1. Can Euler vortices collapse self-similarly without rotation, for some number of vortices? Theorem 1 excludes the regime of a strong vortex with weak tight clusters, and Section 4 finds none along two families up to 79 vortices. For $\alpha = 0$ such a collapse is a relative equilibrium of the imaginary circulations $i\Gamma_j$ [6, Sect. 10], a characterization that might be used.
2. Is $\alpha_\infty = \lim_{N \rightarrow \infty} \alpha^*(N)$ positive, along these families or over all collapses without rotation? A proof that $\alpha^*(N)$ is bounded below by a positive constant would settle the Euler question for the families. Short of that, the thresholds $\alpha^*(N)$ can be certified by the interval methods of [6].
3. Does the continuum problem (15) have a solution, and is its least winding P_∞ a known constant? What does the unread work of O’Neil [13] say about it? Does a sheet with another behavior at its tips, or with more point vortices, wind less?
4. Are the thresholds of Section 4 global minima, that is, are 11 at $\alpha = 2$ and 60 at $\alpha = 1$ the least numbers of vortices that collapse without rotation?
5. Can the expansion (14) be proved, and does a strong vortex with clusters whose e_C are all nonpositive satisfy $P > \sqrt{3}/2$ at every small fixed γ , as pairs do [6, Proposition 4]?

Data availability. The programs `verify_cluster_identities.py`, `verify_cluster_remainders.py`, `verify_cluster_collapses.py`, `verify_cluster_stored.py`, `verify_phase_diagram.py`, `verify_family_structure.py`, `verify_stability.py` and `verify_continuum_limit.py`, with the modules `collapse_core.py`, `continuum_model.py` and `continuum_arb.py`, and `plot_phase_diagram.py`, used for Figure 1, and their output and inputs will be in the repository <https://github.com/ChaseHendrick/collapse-without-rotation>: the programs in `code/`, their inputs and output in `data/`, and the figure in `paper/figures/`.

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